

CISCO Packet Tracer Lab 2 Report

VLAN Configuration & Network Segmentation

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Lab Objective

The objective of this lab was to configure VLANs to segment a network into separate broadcast domains for Students and Faculty. We set up a router-on-a-stick configuration to enable inter-VLAN routing, allowing devices in different VLANs to communicate through router subinterfaces. The lab focuses on creating VLANs on switches, assigning ports to VLANs, configuring trunk ports, setting up router subinterfaces, and verifying connectivity within and between VLANs using ping commands.

Tasks:

1. Configure VLAN 10 (Students) and VLAN 20 (Faculty) on switches.
2. Assign switch ports to the respective VLANs.
3. Set up subinterfaces on the router to enable inter-VLAN routing.
4. Verify VLAN communication by testing connectivity within VLANs and between VLANs.

Equipment and Resources

Hardware:

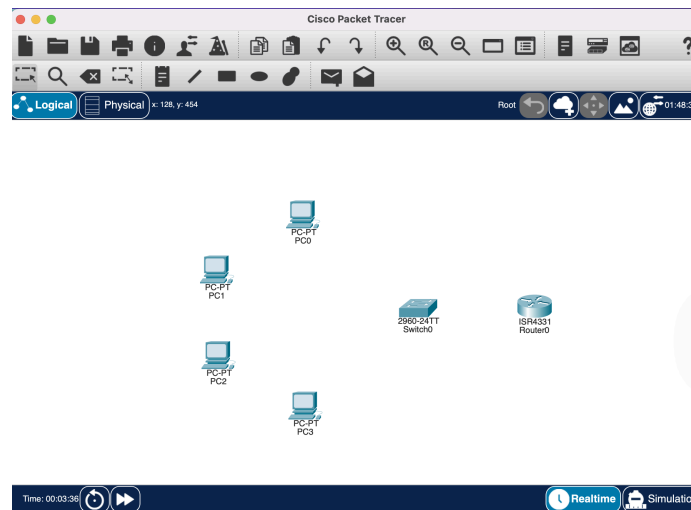
- 1x Cisco ISR4331 Router
- 1x Cisco 2960-24TT Switch
- 4x PC End Devices
- 5x Copper Straight-Through Cables

Software:

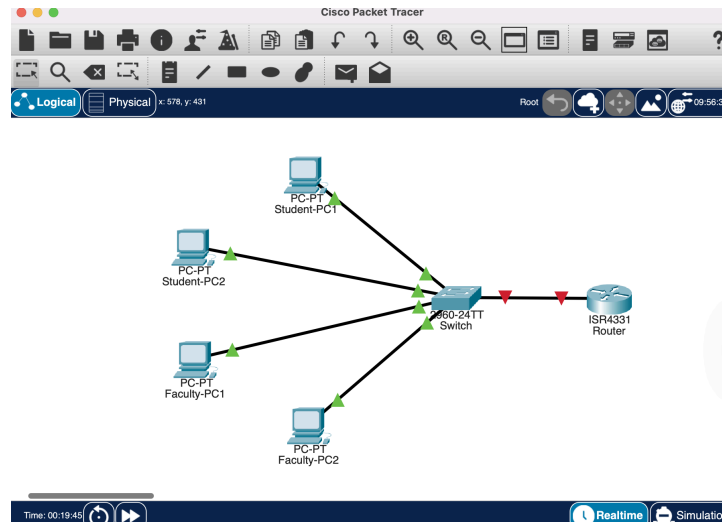
- Cisco Packet Tracer (Network Simulation Software)

Lab Steps

Step 1: Initial Network Topology

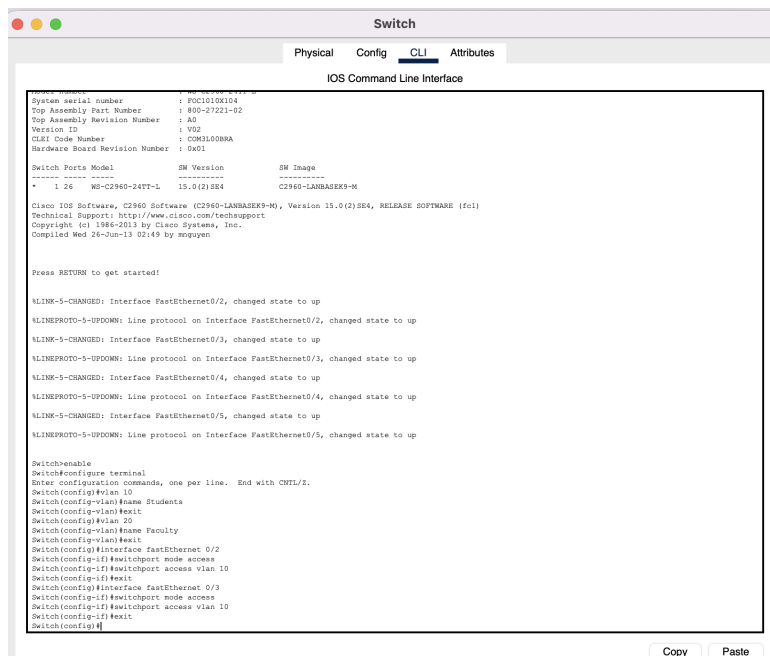


Step 2-3: Connected & Renamed PCs



First, we added one ISR4331 router, one 2960-24TT switch, and four PC end devices. They were arranged in a simple layout where all PCs connect to the switch, and the switch connects to the router via a single trunk link. This created a router-on-a-stick topology. We connected all the devices using Copper Straight-Through cables. Student-PC1, Student-PC2, Faculty-PC1, and Faculty-PC2 were connected to the switch and then the switch was connected to the router. We renamed the PCs for clarity. Student-PC1, Student-PC2 for VLAN 10 and Faculty-PC1, Faculty-PC2 for VLAN 20.

Steps 4-7: VLAN Configuration on Switch



```
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/5
```

```
Switch(config)#interface fastEthernet 0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
```

```
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Next, we accessed the switch through the CLI to create the VLANs. We created VLAN 10 named Students and VLAN 20 named Faculty. Then we assigned ports FastEthernet 0/2 and 0/3 to VLAN 10 using the switchport access vlan 10 command. Ports FastEthernet 0/4 and 0/5 were assigned to VLAN 20. Finally, we configured port FastEthernet 0/1 as a trunk port using switchport mode trunk to carry both VLANs to the router.

Step 8: Verified VLANs with show vlan brief

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 Students	active	Fa0/2, Fa0/3
20 Faculty	active	Fa0/4, Fa0/5
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

To verify our VLAN configuration, we ran the show vlan brief command. The output confirmed that VLAN 10 (Students) had ports Fa0/2 and Fa0/3 assigned, and VLAN 20 (Faculty) had ports Fa0/4 and Fa0/5 assigned. The trunk port (Fa0/1) did not appear in the VLAN list because trunk ports carry multiple VLANs.

Step 9-11: Router Subinterface Configuration

```
IOS Command Line Interface
If you require further assistance please contact us by sending email to
export@cisisco.com.

cisco ISR4331/K9 (1RU) processor with 1795999K/6147K bytes of memory.
Processor board ID FLM23201000
3 Gigabit Ethernet interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3207167K bytes of flash memory at bootflash:.
0K bytes of WebUI ODM Files at webui:.

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Press RETURN to get started!

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

```

Router(config)#interface gigabitEthernet 0/0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Here we accessed the router through the CLI to set up router-on-a-stick. First, we turned on the interface GigabitEthernet0/0/0 using the 'no shutdown' command. Then we created subinterface Gi0/0/0.10 for VLAN 10 with encapsulation dot1Q 10 and assigned it IP address 192.168.10.1. We created subinterface Gi0/0/0.20 for VLAN 20 with encapsulation dot1Q 20 and assigned it IP address 192.168.20.1.

Step 12: Tested with show ip route command

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

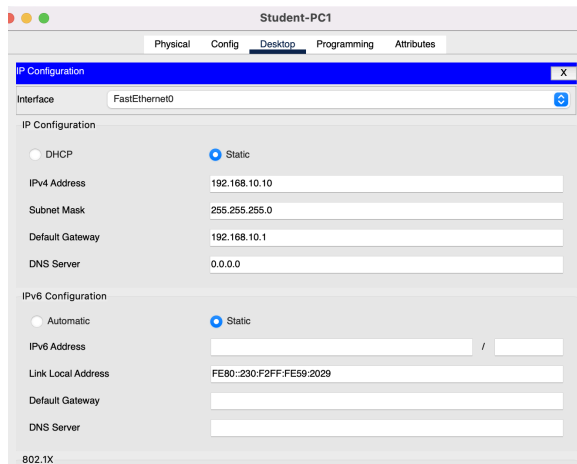
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0.20

Router#

```

To verify the router configuration, we ran the show ip route command. The output showed both VLAN networks as directly connected through the router subinterfaces. This confirmed that the router was ready to route traffic between the two VLANs.

Step 13-16: PC IP Configuration



Student-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

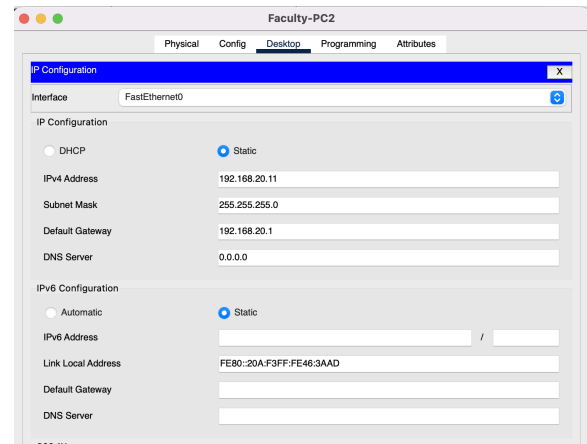
☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::230:F2FF:FE59:2029

Default Gateway

DNS Server



Faculty-PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

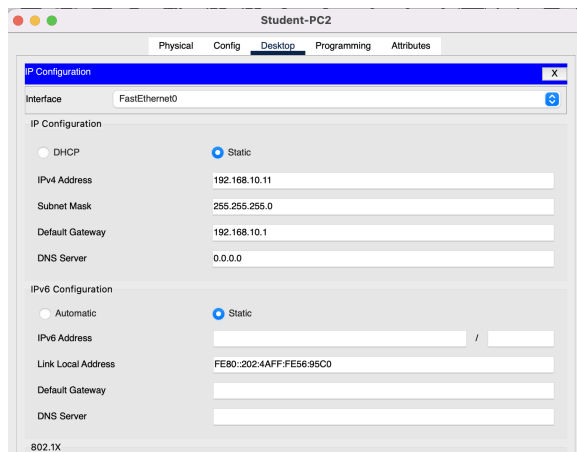
☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::20A:F3FF:FE46:3AAD

Default Gateway

DNS Server



Student-PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

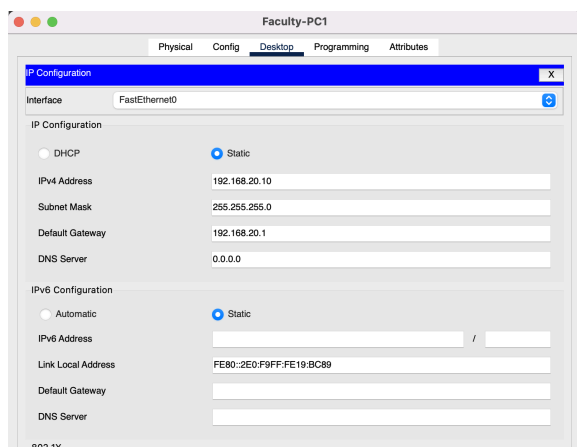
☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::202:4AFF:FE56:95C0

Default Gateway

DNS Server



Faculty-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2E0:F9FF:FE19:BC89

Default Gateway

DNS Server

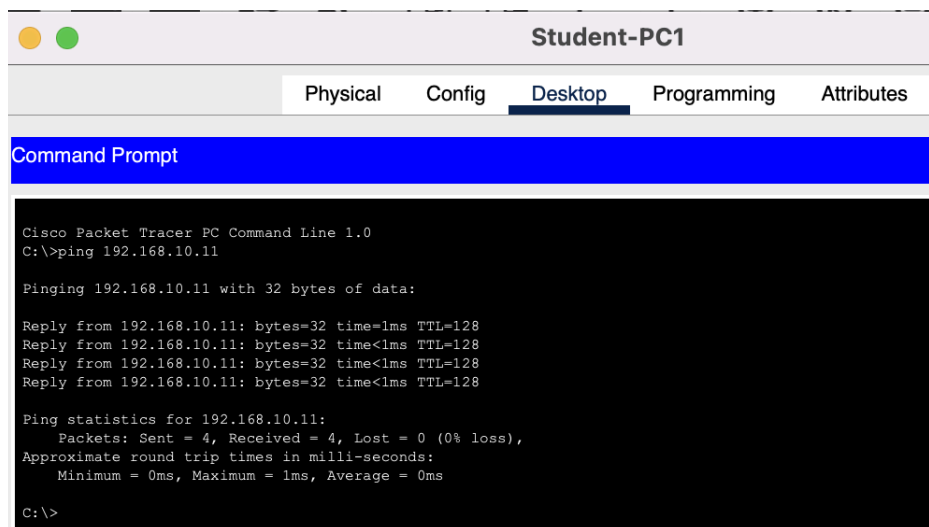
We manually configured each PC with a static IP address by going to the Desktop tab and clicking on IP Configuration. We selected Static instead of DHCP and entered the appropriate IP addresses for each VLAN. Student PCs were given addresses in the 192.168.10.0 network, and

Faculty PCs were given addresses in the 192.168.20.0 network. All PCs had their gateways set to their respective VLAN gateway addresses.

PC IP Table:

Device	IP Address	Subnet Mask	Default Gateway
Student-PC1	192.168.10.10	255.255.255.0	192.168.10.1
Student-PC2	192.168.10.11	255.255.255.0	192.168.10.1
Faculty-PC1	192.168.20.10	255.255.255.0	192.168.20.1
Faculty-PC2	192.168.20.11	255.255.255.0	192.168.20.1

Step 17-18: Connectivity Testing Within VLANs



The screenshot shows the 'Student-PC1' configuration window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the following output:

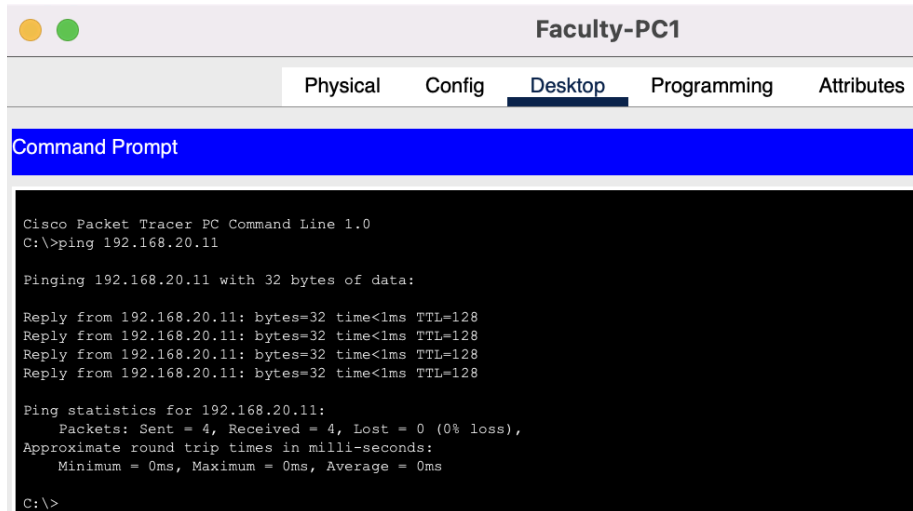
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

The screenshot shows a window titled "Faculty-PC1" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a "Command Prompt" window. The command prompt shows the execution of the command "ping 192.168.20.11". The output indicates that the ping was successful, with 4 replies received, 0% packet loss, and a round trip time of 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

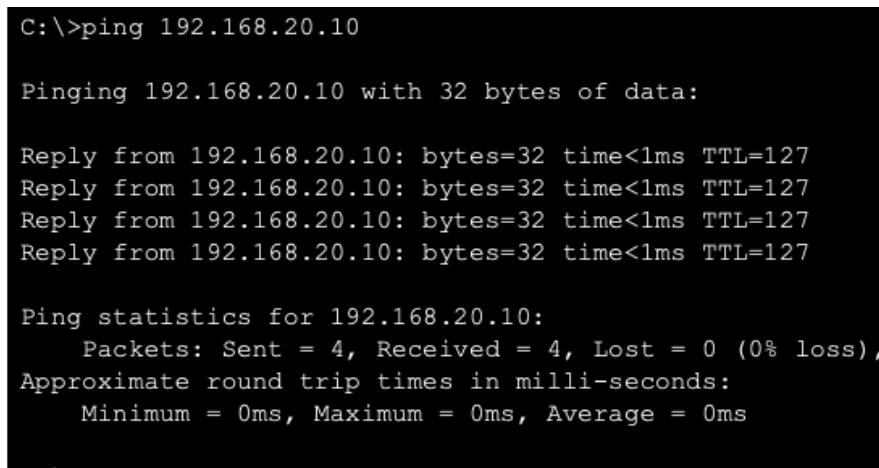
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

To verify connectivity within each VLAN, we used the ping command. From Student-PC1, we pinged Student-PC2 at 192.168.10.11 and received 4 replies with 0% packet loss. From Faculty-PC1, we pinged Faculty-PC2 at 192.168.20.11 and also received 4 replies with 0% packet loss. This confirmed that devices within the same VLAN could communicate with each other.

Step 19: Inter-VLAN Connectivity Testing



The screenshot shows a command prompt window with the command "ping 192.168.20.10". The output indicates that the ping was successful, with 4 replies received, 0% packet loss, and a round trip time of 0ms.

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

For the final test we used the ping command to test inter-VLAN communication. From Student-PC1, we pinged Faculty-PC1 at 192.168.20.10 and got back 4 replies with 0% packet loss. This proved that the router-on-a-stick configuration was working correctly and devices in different VLANs could communicate through the router subinterfaces.

Results and Observations

The lab was completed successfully. Both VLANS were created and configured properly on the switch, with VLAN 10 (Students) assigned to ports Fa0/2, Fa0/3 and VLAN20 (Faculty) assigned to ports Fa/04 and Fa/05. The show VLAN brief command confirmed everything was set up correctly.

The router subinterfaces worked as expected. After configuring Gi0/0/0.10 and Gi0/0/0.20 with the correct dot1Q encapsulation and IP addresses, the show ip route output showed both networks as directly connected, which told us the router was ready to handle traffic between the two VLANs.

Connectivity tests came back clean. Pinging within each VLAN gave 4 out of 4 replies with 0% packet loss, meaning devices in the same VLAN could reach each other without any issues. The inter-VLAN ping from Student-PCI to Faculty-PCI also succeeded with 0% packet loss, which confirmed the router-on-a-stick setup was working correctly.

Overall, the network behaved exactly as expected. The VLANs kept the two groups logically separated while still allowing them to communicate through the router when needed.

Challenges Faced and Solutions

One of the initial challenges we encountered was configuring the trunk port correctly. We initially forgot to set the switchport mode to trunk on port Fa0/1, which meant only one VLAN's traffic was passing through. To solve this issue we verified the trunk configuration using the show interfaces trunk command and properly configured the port with switchport mode trunk.

We also encountered an issue with the router subinterface configuration. We initially forgot to use the encapsulation dot1Q command with the correct VLAN, which prevented the router from recognizing traffic. We fixed this by carefully matching each subinterface's VLAN to the corresponding VLAN on the switch.

A third problem we encountered was the first ping between VLANs showing packet loss. The first packet timed out due to ARP resolution. We learned that this is expected and the following pings showed 0% packet loss confirming the configuration was correct.

Lessons Learned

1. VLANs create logical network separation on the same physical switch. Devices in different VLANs cannot communicate without a router even if they're on the same switch.

2. Router-on-a-stick uses a single physical connection with multiple virtual subinterfaces. Each subinterface needs dot1Q encapsulation matching the VLAN ID.
3. Trunk ports carry traffic for multiple VLANs using VLAN tags. Access ports carry traffic for only one VLAN and remove the tag before forwarding to end devices.
4. The first ping often shows packet loss due to ARP resolution. This is normal and other pings will succeed once devices learn MAC addresses.

Real-World Application

VLANs are often used in real-world networks such as universities, offices, and hospitals to separate different groups of users while using the same physical network. Students, faculty, and administrative staff can all connect to the same switch, but VLANs keep their traffic logically separated which improves network organization, security, and performance by reducing unnecessary broadcast traffic.

Network segmentation also helps protect sensitive information. Departments like finance, administration, or faculty can be placed on separate VLANs where access to their data can be better controlled. Router-on-a-stick configurations are commonly used in small to medium networks because they allow multiple VLANs to communicate through a single router interface while keeping the network flexible and scalable.

Conclusion

In this lab, we had successfully configured VLANs to segment a network into separate groups for students and faculty. We created VLAN 10 and VLAN 20 on the switch, assigned the correct ports, and configured a trunk connection to the router. We then had subinterfaces that were set up using 802.1Q encapsulation to allow communication between VLANs.

After configuring the devices, we made sure the setup using commands like *show vlan* brief and *show ip route* while testing connectivity using ping. The results confirmed that devices within the same VLAN could communicate and that inter-VLAN routing worked correctly. Overall, this lab helped us understand how VLANs and router-on-a-stick configurations are used to be utilized and manage modern networks.